Cybersecurity - Graduate Level Assessment 12 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. How does a buffer overflow on the stack facilitate running attacker-injected code?
   (a) By overwriting the return address to point to the location of that code
   (b) By writing directly to the instruction pointer register the address of the code
   (c) By writing directly to %eax the address of the code
   (d) By changing the name of the running executable, stored on the stack

2. The _______________ was a huge marketplace of Dark Web specifically famous for selling of illegal drugs & narcotics as well as you can find a wide range of other goods for sale.
   (a) Silk Road
   (b) Cotton Road
   (c) Dark Road
   (d) Drug Road

3. Which of the following does authentication aim to accomplish?
   (a) Restrict what operations/data the user can access
   (b) Determine if the user is an attacker
   (c) Flag the user if he/she misbehaves
   (d) Determine who the user is

4. Which of the following is not a transport layer vulnerability?
   (a) Mishandling of undefined, poorly defined variables
   (b) The Vulnerability that allows "fingerprinting" & other enumeration of host information
   (c) Overloading of transport-layer mechanisms
   (d) Unauthorized network access

5. What are the types of scanning?
   (a) Port, network, and services
   (b) Network, vulnerability, and port
   (c) Passive, active, and interactive
   (d) Server, client, and network

True / False (10 marks)

Answer True or False

6. True or False: UNIX and NT-based systems are known to have buffer overflow vulnerabilities due to their handling of memory.

7. True or False: Buffer-overflow issues can be resolved entirely through adequate memory checks alone without the need for boundary checks.

8. True or False: Public key encryption offers significant advantages in secure key exchange compared to symmetric key cryptography.

9. True or False: A packet filter firewall primarily filters traffic at the network or transport layer of the OSI model.

10. True or False: A digital signature relies on a public-key system to ensure authenticity and integrity of digital communications.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the difference between two consecutive numbers in a given list.

12. Write a python function to find the smallest prime divisor of a number.

End of Paper